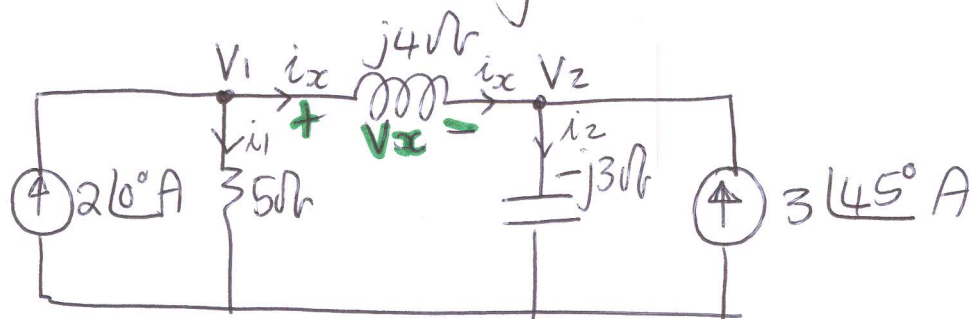


Tutorial 8: EBN III

①

10.16) Nodal analysis: Find V_x ?



1st: Convert circuit to freq. domain: Done ✓

2nd: Nodal analysis: KCL

$$V_1 - V_2 = V_x$$

$$i_x = \frac{V_x}{j4}$$

Node 1: $2\angle 0^\circ = \frac{V_x}{j4} + \frac{V_1}{5}$

$$2\angle 0^\circ = \frac{V_1 - V_2}{j4} + \frac{V_1}{5} \quad \text{--- ①}$$

Node 2: $i_x + 3\angle 45^\circ = i_z$

$$\frac{V_1 - V_2}{j4} + 3\angle 45^\circ = \frac{V_2}{-j3} \quad \text{--- ②}$$

Simplify ① & ②

① $2(j20) = 5(V_1 - V_2) + j4V_1$

$$40j = (5 + j4)V_1 - 5V_2$$

$\div 20j$ $2 = \left(\frac{5}{20j} + \frac{4}{20}\right)V_1 - \frac{5}{20j}V_2$

$$2 = (0.25j + 0.2)V_1 - 0.25jV_2$$

(2)

(2)

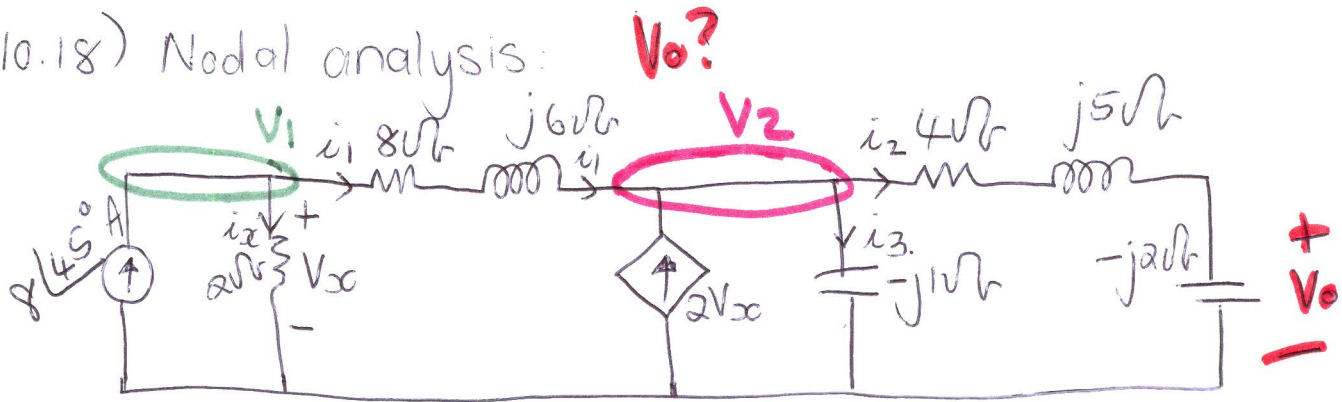
$$\begin{aligned}
 X - (j^2 12): -j3(V_1 - V_2) + (3 \angle 45^\circ)(12) &= j4(V_2) \\
 -j3V_1 + (j3 - j4)V_2 &= -36 \angle 45^\circ \\
 -3jV_1 - jV_2 &= -(25.46 - 25.46j)
 \end{aligned}$$

IN Matrix form

$$\begin{bmatrix} 0.2 + 0.25j & -0.25j \\ 3j & j \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ +25.46 - 25.46j \end{bmatrix}$$

$$V_2 = V_1 - V_2$$

10.18) Nodal analysis:



Frequency domain ✓

$$V_x = V_1$$

KCL node 1:

$$8\angle 45^\circ = i_1 + i_x$$

$$8\angle 45^\circ = \frac{V_1 - V_2}{8 + j6} + \frac{V_1}{2}$$

simplify:

$$2(8 + j6)(8\angle 45^\circ) = 2V_1 - 2V_2 + (8 + j6)V_1$$

$$(16 + 12j)(8\angle 45^\circ) = (10 + j6)V_1 - 2V_2$$

$$22.63 + 158.39j = (10 + j6)V_1 - 2V_2 \quad \text{--- (1)}$$

$$0.1029 + 0.217j$$

KCL node 2:

$$i_1 + 2V_x = i_2 + i_3 \quad ; \quad V_x = V_1$$

$$\frac{V_1 - V_2}{8 + j6} + 2V_1 = \frac{V_2 - 0}{4 + j5 - j2} + \frac{V_2}{-j1}$$

$$\frac{V_1 - V_2}{8 + j6} + 2V_1 = \frac{V_2}{4 + 3j} + jV_2$$

$$\begin{aligned} \times (8 + j6)(4 + 3j): & (4 + 3j)(V_1 - V_2) + (28 + 96j)V_1 = (8 + j6)V_2 + (-48 + 14j)V_2 \\ = 14 + 48j & (32 + 99j)V_1 + (36 - 23j)V_2 = 0 \end{aligned}$$

$$\therefore V_1 = -\frac{(36 - 23j)V_2}{(32 + 99j)}$$

$$V_1 = \left(\frac{45}{433} + \frac{172}{433}j \right) V_2 \quad \text{--- (2)}$$

Substitute ② in ①

④

$$22.63 + 158.39j = (10 + j6) \left(\frac{45}{433} + \frac{172}{433}j \right) V_2 - 2V_2$$

$$V_2 = \frac{22.63 + 158.39j}{-\frac{1448}{433} + \frac{1990}{433}j}$$

$$V_2 = \underline{28.15 \angle -44.17^\circ \text{ V}}$$

$$V_0 = \left(\frac{-j2}{-j2 + j5 + 4} \right) V_2 \quad \dots \quad \text{Voltage division}$$

$$V_0 = (-0.24 - 0.32j) V_2$$

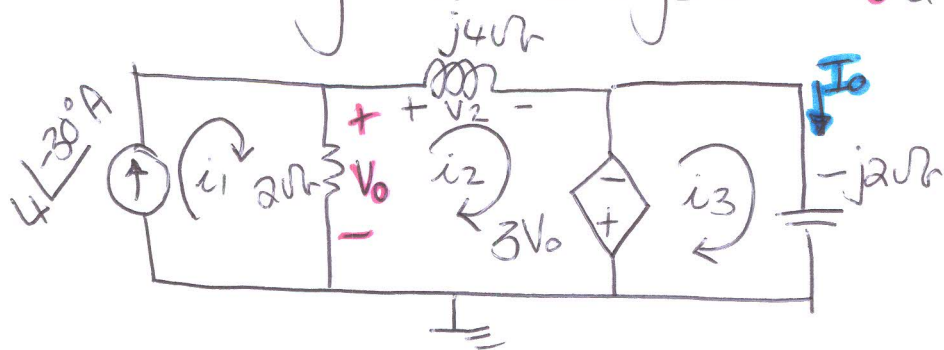
$$V_0 = 11.26 \angle -171.04^\circ \text{ V}$$

+360° For ⊕ angle.

$$\underline{V_0 = 11.26 \angle 188.96^\circ \text{ V}}$$

10.32) Using mesh analysis: V_o & I_o ?

(5)



Apply KVL mesh 1:

$$i_1 = 4 \angle -30^\circ \text{ A}$$

$$= 2\sqrt{3} - 2j \text{ A} \dots \textcircled{1}$$

KVL mesh 2:

$$-V_o + V_2 - 3V_o = 0 \quad \text{But } V_o = \frac{(i_1 - i_2)2}{2}$$

$$-\frac{(i_1 - i_2)2}{2} + (i_2)(j4) - 3(i_1 - i_2)(2) = 0$$

$$-2i_1 + 2i_2 + (j4)i_2 - 6i_1 + 6i_2 = 0$$

$$-8i_1 + (8 + j4)i_2 = 0 \dots \textcircled{2}$$

Substitute ① in ②

$$-8(2\sqrt{3} - 2j) + (8 + j4)i_2 = 0$$

$$i_2 = 1.971 - 2.986j \text{ A}$$

$$\therefore V_o = (2\sqrt{3} - 2j - (1.971 - 2.986j))2$$

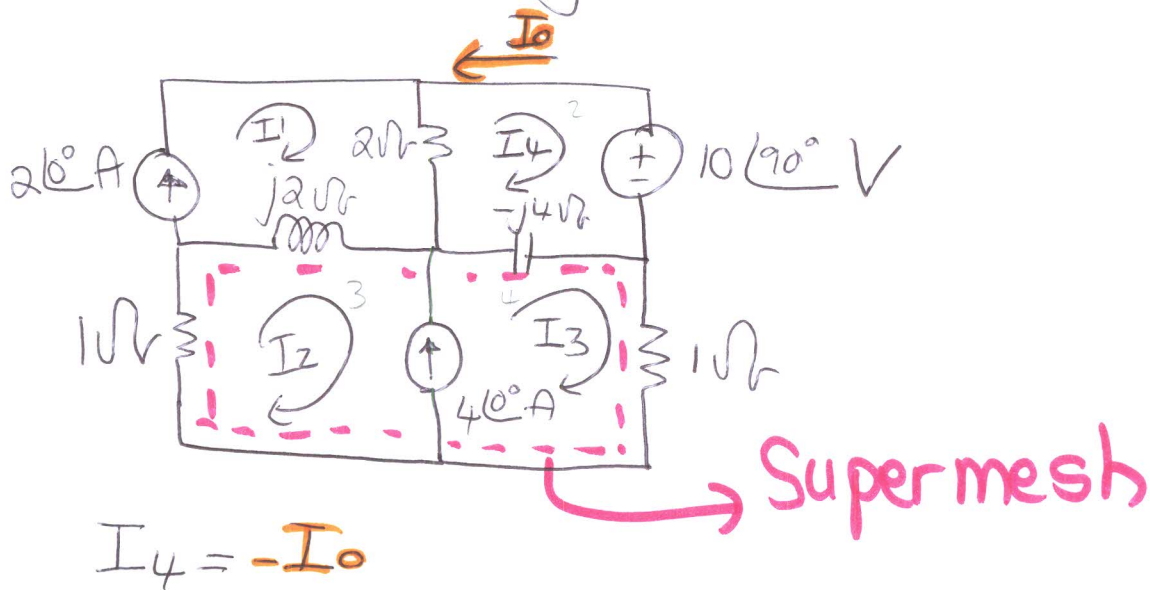
$$\underline{V_o = 3.579 \angle 33.44^\circ \text{ V}}$$

$$I_o = \frac{(-3V_o)}{-j2} = \frac{3V_o}{j2} = \frac{3(3.579 \angle 33.44^\circ)}{j2}$$

$$\underline{I_o = 5.368 \angle -56.56^\circ \text{ A}}$$

10.38) Use mesh analysis. Find I_o ?

(6)



KVL mesh 1: $I_1 = 2\angle 0^\circ \text{ A} \dots \textcircled{1}$

KVL Supermesh $(1)I_2 + j2(I_2 - I_1) - j4(I_3 - I_4) + 1I_3 = 0 \dots \textcircled{2}$

and: $I_3 - I_2 = 4\angle 0^\circ \text{ A} \dots I_2 = I_3 - 4\angle 0^\circ$

KVL mesh 4: $10\angle 90^\circ - j4(I_4 - I_3) + 2(I_4 - I_1) = 0 \dots$

But: $I_1 = 2\angle 0^\circ \text{ A}$ $10\angle 90^\circ + (2 - j4)I_4 - 2(2\angle 0^\circ) + j4I_3 = 0$
 $10.77\angle 111.80^\circ + (j4)I_3 + (2 - j4)I_4 = 0$
 $(j4)I_3 + (2 - j4)I_4 = 4 - 10j$

Substitute $\textcircled{1}$ in $\textcircled{2}$ and $I_2 = I_3 - 4\angle 0^\circ$

$$I_2(1 + j2) - j2(2\angle 0^\circ) + (1 - j4)I_3 + j4I_4 = 0$$

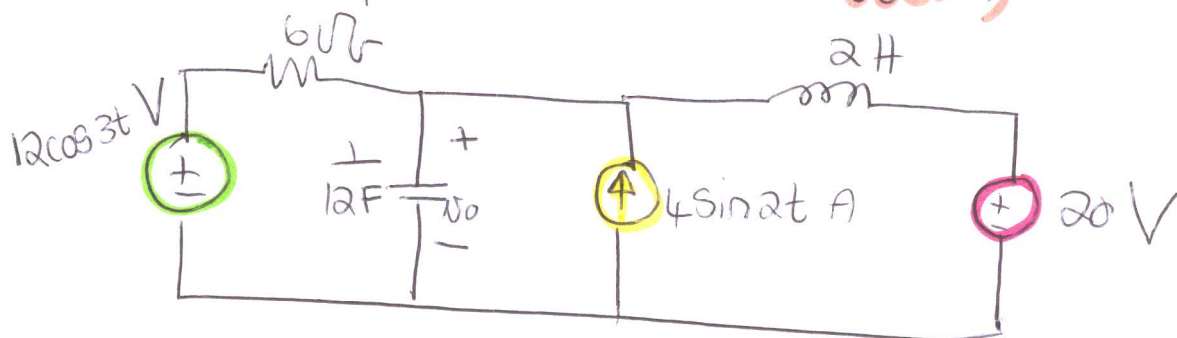
$$(I_3 - 4\angle 0^\circ)(1 + j2) + 4\angle -90^\circ + (1 - j4)I_3 + j4I_4 = 0$$

$$12.65\angle -108.43^\circ + (2 - 2j)I_3 + j4I_4 = 0$$

$$\begin{bmatrix} j4 & 2 - j4 \\ 2 - 2j & j4 \end{bmatrix} \begin{bmatrix} I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 4 - 10j \\ 4 + 12j \end{bmatrix} \quad \text{Can simplify more.}$$

10.46) Use Superposition: Find $v_o(t)$

(7)



Three different frequencies.

$$v_o = v_1 + v_2 + v_3$$

$v_1 \rightarrow$ DC voltage source $\omega = 0$

ac Voltage source $v_2 \rightarrow 12\cos(3t) \text{ V}, \omega = 3$

ac Current $v_3 \rightarrow 4\sin(2t) \text{ A}, \omega = 2$

* When Source = 0

Current source: open circuit

Voltage source: short circuit

For v_1 :

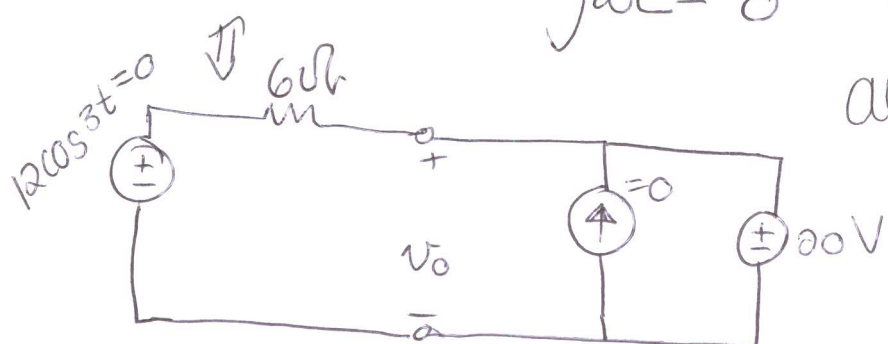
$$\omega = 0; \frac{1}{j\omega C} = \infty$$

(open-circuit)

$$j\omega L = 0$$

(short-circuit)

all other source $\equiv 0$, DC = 20 V

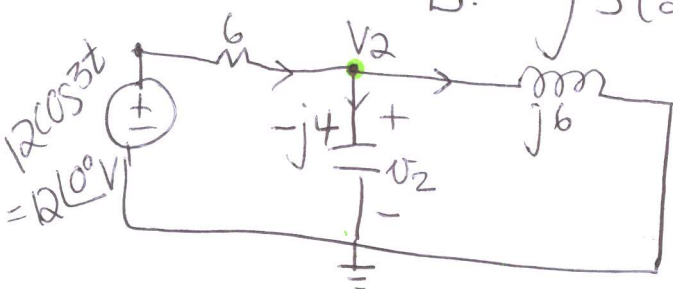


$$v_o = v_1 = 12 \text{ V}$$

For v_2 :

$$\omega = 3; \frac{1}{j3(\frac{1}{12})} = -j4 \Omega = Z_C$$

$$L: j3(2) = j6 \Omega = Z_L$$



KCL v_2 :

$$\frac{12\angle 0^\circ - v_2}{6} = \frac{v_2}{-j4} + \frac{v_2}{j6}$$

$$\times 6(-j4)(j6)$$

$$= 144$$

$$24(12\angle 0^\circ - v_2) = 36jv_2 + 24jv_2$$

$$288 - 24v_2 = 12jv_2$$

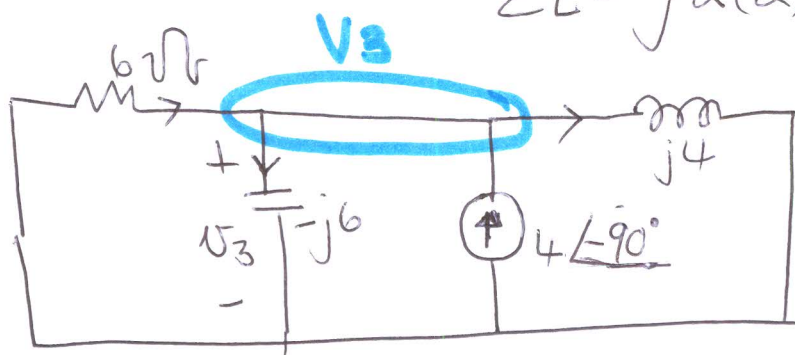
$$v_2(24 + 12j) = 288$$

$$v_2 = 10.733 \angle -26.57^\circ \text{ V}$$

(8)

For V_3 : $\omega = 2$; $Z_C = \frac{1}{j2(\frac{1}{12})} = -j6 \Omega$.

$$Z_L = j2(2) = j4 \Omega$$



$$\begin{aligned} & 4 \sin 2t \\ &= 4 \cos(2t - 90^\circ) \\ &= 4 \angle -90^\circ \text{ A} \end{aligned}$$

KCL V_3 :

$$\frac{-V_3}{6} + 4 \angle -90^\circ = \frac{V_3}{-j6} + \frac{V_3}{j4}$$

$$\times 6(-j6)(j4): -24V_3 + 576 \angle -90^\circ = 6(j4)V_3 + 6(-j6)V_3$$

$$\begin{aligned} &= 6 \times 24 \\ &= 144 \end{aligned}$$

$$26.83 \angle -26.57^\circ V_3 = 576 \angle -90^\circ$$

$$V_3 = 21.47 \angle -63.43^\circ \text{ V}$$

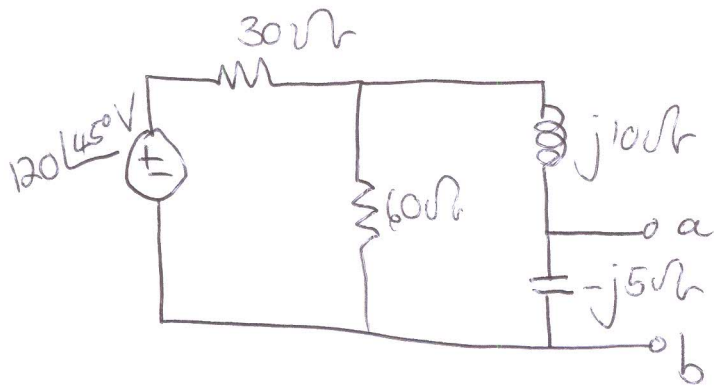
$$v_o(t) = 12 + 10.733 \cos(3t - 26.57^\circ) + 21.47 \cos(2t - 63.43^\circ) \text{ V}$$

transfer to Sin
+ 90°

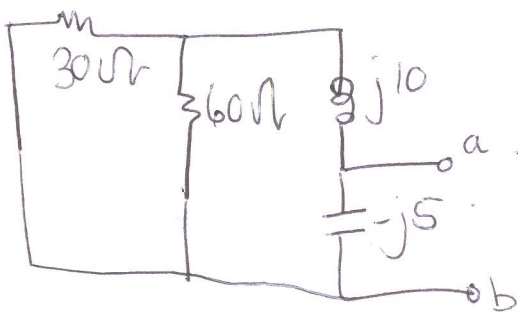
$$v_o(t) = 12 + 10.733 \cos(3t - 26.57^\circ) + 21.47 \sin(2t + 26.57^\circ) \text{ V}$$

10.56 b) Obtain Thevenin & Norton equivalent

⑨

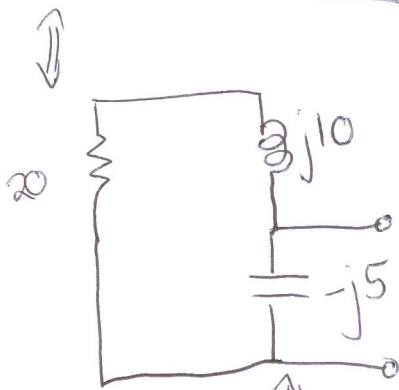


Thevenin: $R_{th} = Z_{th} = Z_N$ Voltage Source = 0
+Norton

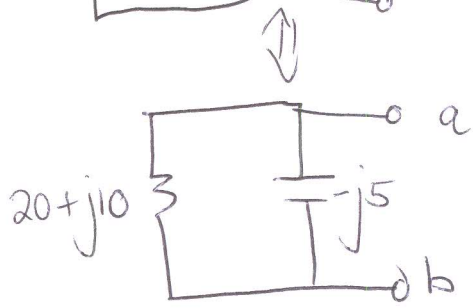


$$\equiv 30 \parallel 60$$

$$Z_{p1} = \frac{(30)(60)}{30+60} = 20 \Omega$$



$$\equiv j10 + 20 \text{ in series} = Z_{s1}$$

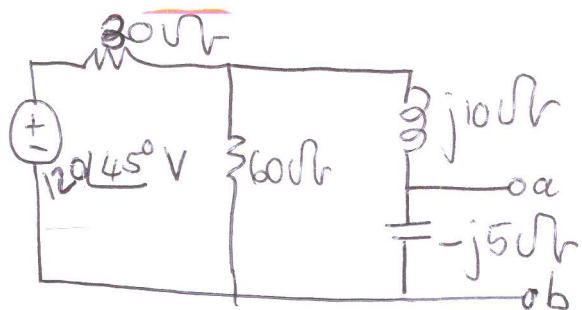


$$Z_{th} = \frac{(20+j10)(-j5)}{20+j10-j5} = Z_N$$

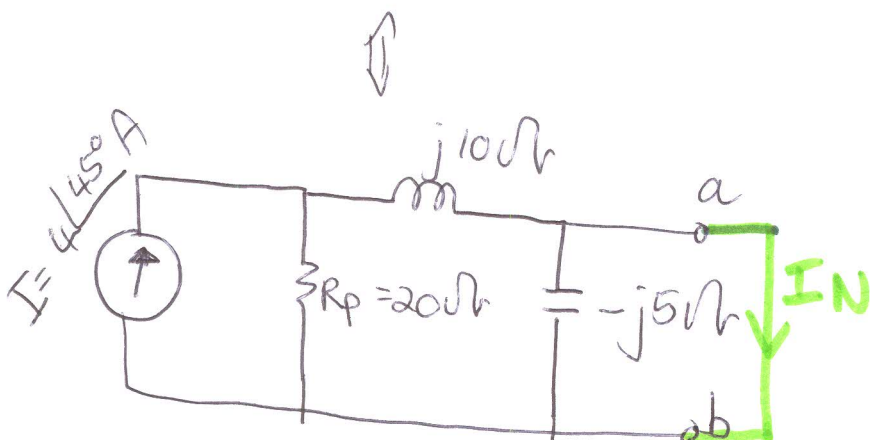
$$= \frac{50 - 100j}{20+j5}$$

$$= 1.176 - 5.29j \Omega$$

$$= 5.423 \angle -77.47^\circ \Omega$$



Firstly: Transform voltage source to current source & combine parallel resistors.



$$R_p = 60 \parallel 30 = 20\Omega$$

$$I = \frac{V}{R} = \frac{120\angle 45^\circ}{30} = 4\angle 45^\circ \text{ A}$$

For I_N :

$$I_N = \frac{20}{20 + j10} (4\angle 45^\circ) \dots \text{Current division}$$

$$= 3.578 \angle 18.43^\circ \text{ A}$$

$$\therefore \underline{V_{th}} = I_N Z_{th}$$

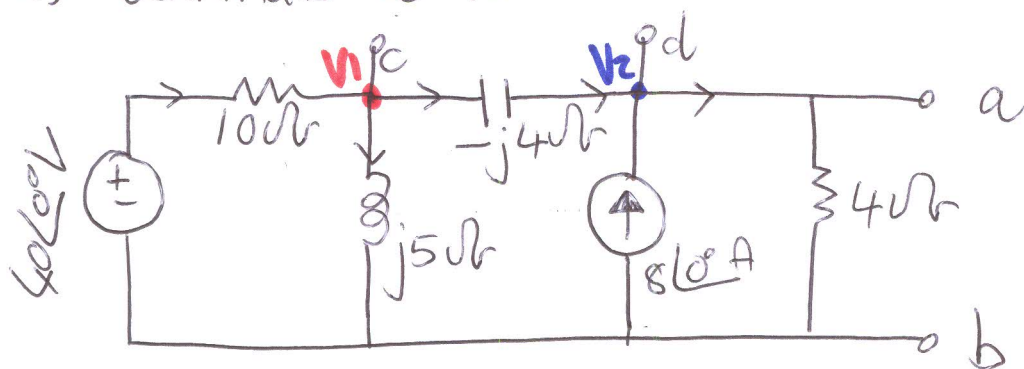
$$= (3.578 \angle 18.43^\circ) (5.423 \angle -77.47^\circ)$$

$$= 19.4 \angle -59.04^\circ \text{ V}$$

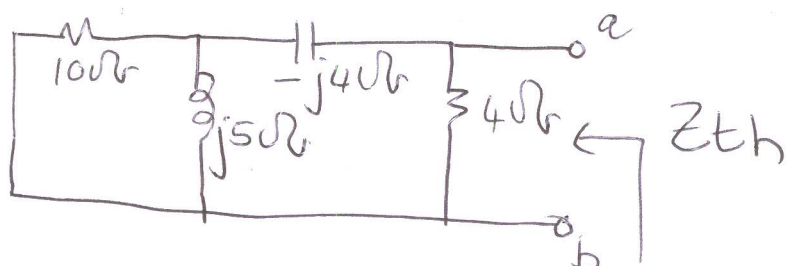
10.60) Find Thevenin equivalent

(11)

a) terminals a-b



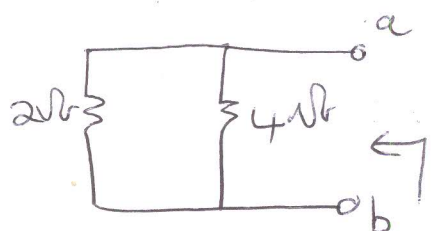
Z_{th} : Sources are zero!



$$10 \parallel j5 = \frac{(10)(j5)}{10 + j5} = \frac{2 + 4j\Omega}{1} \rightarrow$$

↓
in series with $-j4$

$$= 2\Omega$$



$$\begin{aligned} Z_{th} &= 2 \parallel 4 \\ &= \frac{4}{3} \Omega \\ &= 1.333 \Omega \end{aligned}$$

→

For V_{th} : $V_2 = V_{th}$

KCL node 1: $\frac{40\angle 0^\circ - V_1}{10} = \frac{V_1}{j5} + \frac{V_1 - V_2}{-j4}$

$$(x(10)(j5)(-j4) = 200) \quad 20(40\angle 0^\circ - V_1) = -j40V_1 + j50(V_1 - V_2)$$

$$800 - 20V_1 + j40V_1 - j50V_1 + j50V_2 = 0 \quad (12)$$

$$(-20 + j40 - j50)V_1 + j50V_2 = -800$$

$$(20 + j10)V_1 - j50V_2 = 800$$

$$(1 + j0.5)V_1 - 2.5jV_2 = 40 \quad \text{----- } (1)$$

KCL Node 2

$$\frac{V_1 - V_2}{-j4} + 8\angle 0^\circ = \frac{V_2}{4}$$

$$(X - 16j) \quad 4(V_1 - V_2) + (-16j)(8) = -j4V_2$$

$$4V_1 - 4V_2 - 128j + j4V_2 = 0$$

$$4V_1 = 128j + (4 - j4)V_2$$

$$V_1 = 32j + (1 - j)V_2 \quad \text{----- } (2)$$

Substitute (2) in (1)

$$(1 + j0.5)(32j + V_2 - jV_2) - 2.5jV_2 = 40$$

$$-16 + 32j + V_2 + j0.5V_2 + \frac{1}{2}V_2 - jV_2 - 2.5jV_2 = 40$$

$$(1.5 - 3j)V_2 = 56 - 32j$$

$$V_2 = \underline{19.23\angle 33.69^\circ \text{ V}}$$

$$= \underline{V_{th}}$$

b) terminals c-d

$$10\parallel j5 = 2 + j4$$

↓
in series with 4Ω

$$\equiv 6 + j4\Omega$$

$$\therefore \underline{Z_{th}} = (6 + j4)\parallel -j4 = \underline{2.667 - 4j\Omega}$$

For V_{th}

(13)

use results of part (a) V_1 & V_2

$$V_{th} = V_1 - V_2$$

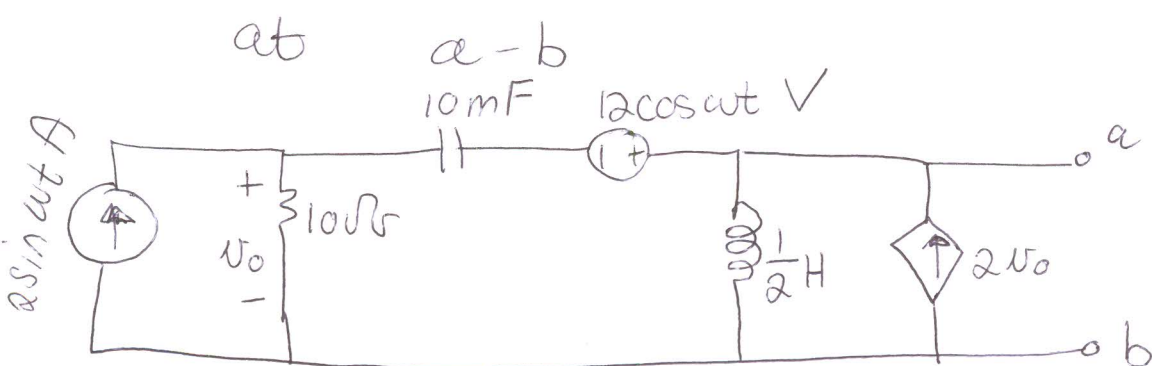
$$V_1 = 32j + (1-j)(19.23 \angle 33.69^\circ)$$

$$V_1 = 37.71 \angle 45^\circ \text{ V}$$

$$\begin{aligned} \therefore V_{th} &= 37.71 \angle 45^\circ - 19.23 \angle 33.69^\circ \\ &= 19.23 \angle 56.31^\circ \text{ V} \end{aligned}$$


10.66) Determine Thevenin & Norton

(14)

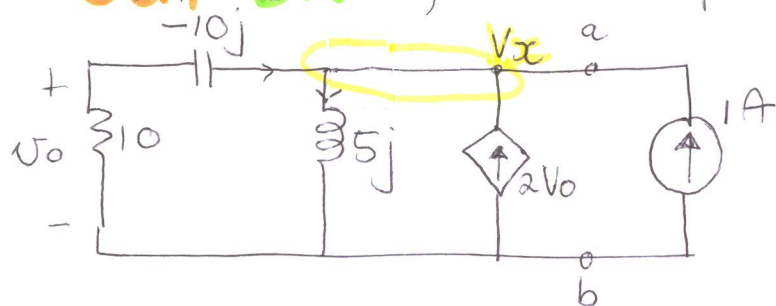


$\omega = 10$; given

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(10)(10 \times 10^{-3})} = -j10 \Omega$$

$$Z_L = j\omega L = j(10)\left(\frac{1}{2}\right) = 5j\Omega$$

For $Z_{th} = Z_N$; all independent source = 0



$V_x = (1) Z_{th}$

$V_o = \frac{10}{10 - 10j} V_x$... ① Voltage division

KCL V_x :

$$\frac{V_o - V_x}{-10j} + 2V_o + 1 = \frac{V_x}{5j}$$

$(\times (-10j)(5j))$: $5j(V_o - V_x) + 100V_o + 50 = -10jV_x$

$(5j + 100)V_o + 5jV_x = -50$ ②

Substitute ① in ②

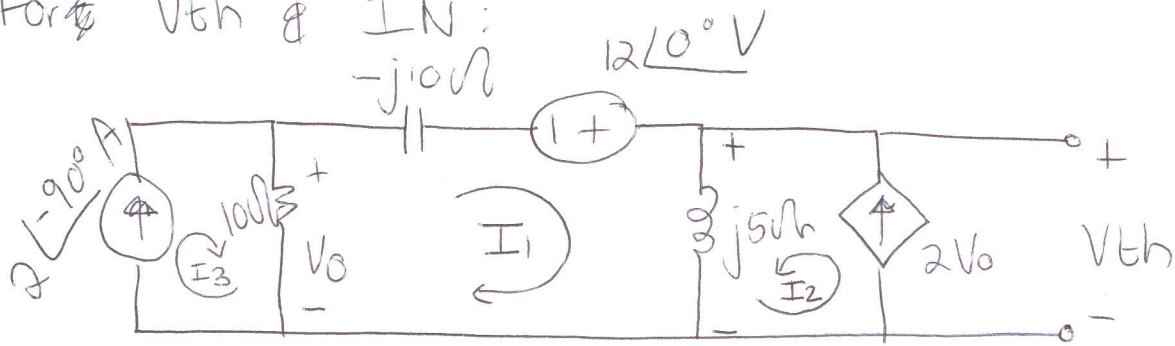
$$(5j + 100)\left(\frac{10}{10 - 10j}\right)V_x + 5jV_x = -50$$

$(47.5 + 57.5j)V_x = -50$ $= Z_N$

$V_x = 0.67 \angle 129.56^\circ V = Z_{th}$

For V_{th} & I_N :

(15)



Find V_o with:

$$I_2 = 2V_o \quad \text{--- (2)}$$

$$\text{KVL } I_1: -12\angle 0^\circ + j5(I_1 + I_2) - V_o - j10I_1 = 0 \quad \text{--- (1)}$$

$$I_3 = 2\angle -90^\circ$$

$$\begin{aligned} V_o &= 10(I_3 - I_1) \\ &= 10(2\angle -90^\circ - I_1) \\ &= 20\angle -90^\circ - 10I_1 \quad \text{--- (3)} \end{aligned}$$

Substitute (2) & (3) in (1)

$$\begin{aligned} -12\angle 0^\circ + j5(I_1 + 2(20\angle -90^\circ - 10I_1)) - 20\angle -90^\circ + 10I_1 - j10I_1 &= 0 \\ -12\angle 0^\circ + j5I_1 + j10(20\angle -90^\circ) - j100I_1 - 20\angle -90^\circ + (10 - j10)I_1 &= 0 \\ (10 - 105j)I_1 &= -188 - 20j \\ I_1 &= 1.792\angle -89.37^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} 2V_o &= 2(20\angle -90^\circ - 10(I_1)) \\ &= 4.17\angle -95.44^\circ \end{aligned}$$

$$\begin{aligned} V_{th} &= (I_2 + I_1)5j \\ &= (2V_o + I_1)5j \\ &= 29.78\angle -3.62^\circ \end{aligned}$$

$$\begin{aligned} I_N &= \frac{V_{th}}{Z_{th}} \\ &= 44.44\angle -133.18^\circ \text{ A} \end{aligned}$$